CASE STUDY

Cost–Performance Evaluation of Internet Connectivity Technologies for Multi-Building Enterprise Networks

COMPUTER COMMUNICATION NETWORKS (BECE406L)

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SUMMARY

Modern organizations operating across geographically distributed campuses require reliable, high-speed connectivity to support essential functions such as cloud-based services, real-time video conferencing, and secure data backups. This study evaluates the effectiveness of five primary connectivity technologies, specifically Digital Subscriber Line (DSL), Business Broadband, Dedicated Leased Lines, Dark Fiber, and Metro Ethernet, to resolve existing network bottlenecks within a mid-sized consultancy firm. The analysis specifically addresses critical failures in the firm's current patchwork infrastructure, including unstable VPN connections at the main headquarters and severe upload constraints at its development centre

To provide an evidence-based recommendation, the study employs a structured, multi-phase methodology grounded in established network engineering theory. Theoretical capacity was calculated using Shannon's Channel Capacity Theorem to determine the physical limits of copper-based mediums, while the Total Network Delay model was applied to evaluate end-to-end latency. These technical performance metrics were then integrated with a 36-month Total Cost of Ownership (TCO) model. Finally, a Multi-Criteria Decision Analysis (MCDA) was used to rank each technology based on its composite Cost-Performance Ratio.

The results demonstrate that while legacy copper technologies are cost-effective, they are fundamentally unsuitable for enterprise use due to exponential signal degradation over campus distances. The analysis concludes that Metro Ethernet represents the most balanced solution, offering high throughput and low latency while simplifying network management into a unified logical LAN.

INTRODUCTION

Background

The internet has evolved into an essential utility for modern enterprise operations. Much like electrical or water grids, a business simply cannot function without reliable connectivity. However, when an organization operates across multiple buildings distributed throughout a city, provisioning adequate internet access becomes a complex engineering challenge that extends far beyond selecting a standard commercial plan from an ISP website.

Enterprises typically evaluate several distinct transmission technologies for this purpose, including legacy telephone lines, shared broadband, and dedicated leased lines. These technologies have existed for varying lengths of time and operate on fundamentally different physical principles. Digital Subscriber Line technology utilizes existing copper telephone wires, offering the advantage of ubiquitous infrastructure and rapid installation. Unfortunately, copper wire was not designed for high-speed data transmission; it suffers from severe signal attenuation over distance, meaning the connection quality degrades significantly the further a building is located from the provider's exchange.

Broadband technologies, such as coaxial cable or Fibre-to-the-Cabinet, improved upon this by delivering much higher baseline speeds. Nevertheless, broadband relies on a shared infrastructure model where a single connection pool is distributed among numerous local subscribers. This shared contention ratio inevitably causes speeds to drop drastically during peak operational hours. At the premium end of the spectrum, leased lines provide a private, dedicated fibre optic connection that is entirely uncontended. These lines guarantee symmetric upload and download speeds and are governed by strict Service Level Agreements that ensure maximum uptime. Ultimately, there is no single universal solution for a multi-building enterprise, as the optimal technology depends heavily on the specific workloads, geographic constraints, and budgetary limitations of each individual facility.

Importance

Selecting network infrastructure based solely on the lowest initial cost is a flawed strategy. In practice, choosing an inadequate technology introduces severe operational bottlenecks that ultimately cost an enterprise far more in lost productivity than the money saved on monthly bills. For instance, if a central headquarters hosting a VPN server relies on an asymmetric broadband connection with limited upload capacity, remote employees will experience frozen video calls, stalled file transfers, and frequent disconnections during peak hours. Similarly, a research and development team that needs to frequently push large code repositories to the cloud will find their daily workflows actively disrupted if their facility is constrained by a distant, low-bandwidth DSL line

Provisioning an expensive dedicated leased line for a small remote office that only requires basic web browsing and email access represents a highly inefficient allocation of corporate funds. The critical importance of this study lies in mitigating these risks. The cost of implementing the wrong infrastructure is substantially higher than the analytical effort required to design it correctly. By utilizing established telecommunications theory and mathematical models, this study ensures that the connectivity recommendation for each building is justified by empirical evidence rather than administrative guesswork.

Objectives

- **Technological Evaluation**: The first objective is to understand the technical mechanisms, inherent limitations, and optimal operational use cases for various connectivity solutions, spanning from legacy DSL to advanced Metro Ethernet and leased lines.
- **Quantitative Performance Modellin**g: The secondary goal is to apply theoretical frameworks, including Shannon's Channel Capacity formula and comprehensive network delay models, to calculate realistic performance metrics based on the specific geographic distances of the enterprise campus.
- **Operational Alignment and Network Design:** The study aims to systematically analyse the distinct daily workloads of each facility and match those operational requirements against the appropriate physical layer technology. Finally, the study will formulate and justify a cost-effective, high-performance recommendation that connects the separate buildings into a cohesive, future-proof network architecture.

PROBLEM STATEMENT

The Situation

A mid-sized consultancy firm operates across three distinct office buildings situated within the same city. Historically, internet connectivity was procured for each site independently, relying on whatever plans were locally available and affordable at the time of deployment. Because no deliberate technical coordination took place, the enterprise now suffers from an increasingly inefficient network architecture as its operations scale.

The infrastructure is divided into three specific operational zones. Building A acts as a high-density primary office housing 100 employees. This location is responsible for managing a central VPN server utilized by roughly 30 remote workers daily, executing a nightly 50 GB offsite cloud backup, and serving as the primary hub for internal video conferencing. Building B functions as the R&D Development Centre, accommodating 50 software engineers. Their workflows are highly data-intensive, requiring constant interaction with Git repositories, automated CI/CD pipelines, and cloud-based container registries. Robust symmetric upload speed is a critical requirement for this facility due to the massive volume of outbound data. Finally, Building C operates as a smaller remote office with 20 employees, where utilization is focused on standard communication tasks such as cloud-based CRM tools, emails, and web browsing.

Physically, Building B is situated approximately 500 meters from the core network and is accessible via an underground conduit. Building C is located 3.5 kilometres away along public roads. This distance metric critically impacts the viability of copper-based technologies due to signal attenuation; specifically, the Shannon Capacity model indicates that VDSL2 technology becomes infeasible beyond approximately 3.67 kilometres.

What is Currently Going Wrong

The primary operational hub at Building A is currently utilizing a standard business broadband connection. During peak business hours, specifically the late morning and early afternoon, the connection becomes entirely saturated by the combined traffic of in-office staff and remote VPN users. This severe network congestion forces VPN sessions to drop and regularly causes the critical nightly cloud backup jobs to fail or bleed into the next operational day. Simultaneously, the developers in Building B are severely constrained by an aging ADSL connection. Pushing code to remote repositories is unacceptably slow, and automated build pipelines take drastically longer than they should, creating a massive bottleneck in the development lifecycle.

At the remote facility in Building C, the basic DSL line lacks any formal SLA. While the baseline bandwidth is generally sufficient for administrative needs, the connection has

suffered three unmitigated outages in the past six months without any guaranteed repair window. Compounding these localized issues is a fundamental architectural flaw; because the buildings are not interconnected via a private wide area network, simple internal file transfers between Building A and Building B must route externally across the public internet, introducing unnecessary latency and routing overhead.

Why This Needs to Be Fixed

- The fundamental issue driving these failures is that the connectivity infrastructure was provisioned without any rigorous technical analysis. Consequently, the enterprise is burdened with mismatched technologies that completely fail to support the actual data demands of each site. The network bottlenecks experienced at the primary office and the R&D centre are actively disrupting core business functions. Unstable VPN connections and failing backups pose direct threats to data security and remote operational continuity. Furthermore, heavily constrained upload speeds at the development centre artificially inflate process times, directly translating to lost engineering productivity.
- From a financial perspective, the organization currently lacks the necessary data to determine if its telecommunications budget is being optimized. The company may be overpaying for underperforming services while simultaneously underspending on critical infrastructure, generating hidden costs across the business. As the enterprise continues to migrate essential services and data storage to the cloud, reliable internet connectivity is rapidly becoming its most critical piece of infrastructure. Continuing to operate on a fragmented, unplanned network architecture while undergoing this transition represents an unacceptable operational risk that the company cannot afford to ignore.

THEORETICAL BACKGROUND

Relevant CCN Concepts

Before deciding on the physical infrastructure for a multi-building network (usually called a Campus Area Network or CAN), we have to establish baseline performance metrics that will dictate our technology choices. Several concepts apply to this scenario:

- **Bandwidth vs. Throughput:** Bandwidth represents the theoretical maximum capacity of a link, whereas throughput is the actual, real-world data transfer rate we can expect after accounting for protocol overhead, network congestion, packet loss, weather conditions, etc.
- **Latency, Jitter, and Quality of Service (QoS):** Latency is the total time it takes for a packet to travel from the source to destination, while jitter is the variance in this latency. For a company running real-time services like video conferencing between buildings, maintaining a low latency and strict jitter control is far more important than raw bandwidth being available.
- **Contention Ratios:** This concept is especially relevant when evaluating broadband. It refers to the number of subscribers sharing the same ISP infrastructure. A high contention ratio means speeds will drop drastically during peak business hours, which can be risky for enterprise networking.
- **Symmetric vs. Asymmetric Routing:** Most commercial internet connections are asymmetric, meaning they offer massive download speeds but very limited upload speeds. However, for a central company building hosting internal servers or pushing daily backups to the cloud, symmetric connections (equal upload and download) are essentially mandatory.

Technology Overview

To bridge the physical gaps between the buildings, several transmission technologies were evaluated for this case study, ranging from standard commercial offerings to advanced enterprise topologies:

1. Telephone Lines (DSL - Digital Subscriber Line): DSL utilizes existing copper telephone infrastructure. Modern iterations like VDSL (Very-high-bit-rate DSL) can offer decent speeds, but they are limited by the physics of a simple copper wire. The signal suffers from severe attenuation over distance and if a company building is located more than a few kilometres away from the ISP's provider location, the available bandwidth drops to near-unusable levels.

2. Broadband (Cable and Fiber-to-the-Cabinet): Business broadband usually comes in two forms. Cable broadband uses coaxial cables and DOCSIS technology, offering high

speeds but suffering heavily from shared contention ratios. Fiber-to-the-Cabinet (FTTC) runs high-speed optical fibre to a street cabinet, then uses copper for the final stretch to the building. This is often the "middle ground" option, providing a good balance of cost and performance, though it still lacks the strict performance guarantees of a private line.

3. Leased Lines: A leased line is a dedicated, uncontended, point-to-point fibre optic connection. Because it is not shared with any other businesses, it provides guaranteed symmetric speeds and comes with strict Service Level Agreements (SLAs) dictating 99.99% uptime, with the obvious drawback being extreme monthly cost.

4. Dark Fiber: This involves leasing unused, "unlit" fibre optic cables already buried by telecom companies. The company attaches its own optical transceivers, gaining virtually unlimited, unshared bandwidth and ultimate security, though it requires expensive in-house hardware.

5. Metro Ethernet (Carrier Ethernet): This technology connects geographically separated buildings across a city using standard Ethernet protocols. To the IT team, it makes the wide area network look and behave exactly like a simple local area network (LAN). It is highly scalable and simplifies routing significantly compared to managing multiple separate leased lines.

Definitions and Equations

To evaluate these networking options, we rely on standard CCN mathematical models.

1. Shannon Channel Capacity Theorem: When evaluating the viability of copper-based DSL or Coaxial links, we have to consider the maximum theoretical data rate of a noisy channel. Shannon's capacity dictates this limit:

$$C = B\,log_2(1 + SNR)$$

Where:

C = Capacity in bits per second

B = Bandwidth of the channel in Hertz

SNR = Signal-to-Noise Ratio.

This equation proves why DSL speeds drop over distance: as the cable gets longer, the noise increases and the signal weaken, lowering the SNR and subsequently destroying the link's capacity.

2. Total Network Delay: When sending data between the main headquarters and satellite buildings, the total latency experienced by the end-user is the sum of four different delays:

$$Total\ Delay\ =\ D_{prop} + D_{transmission} + D_{queue} + D_{proc}$$

Where:

D_{prop} = Propagation Delay (Distance divided by speed of medium)

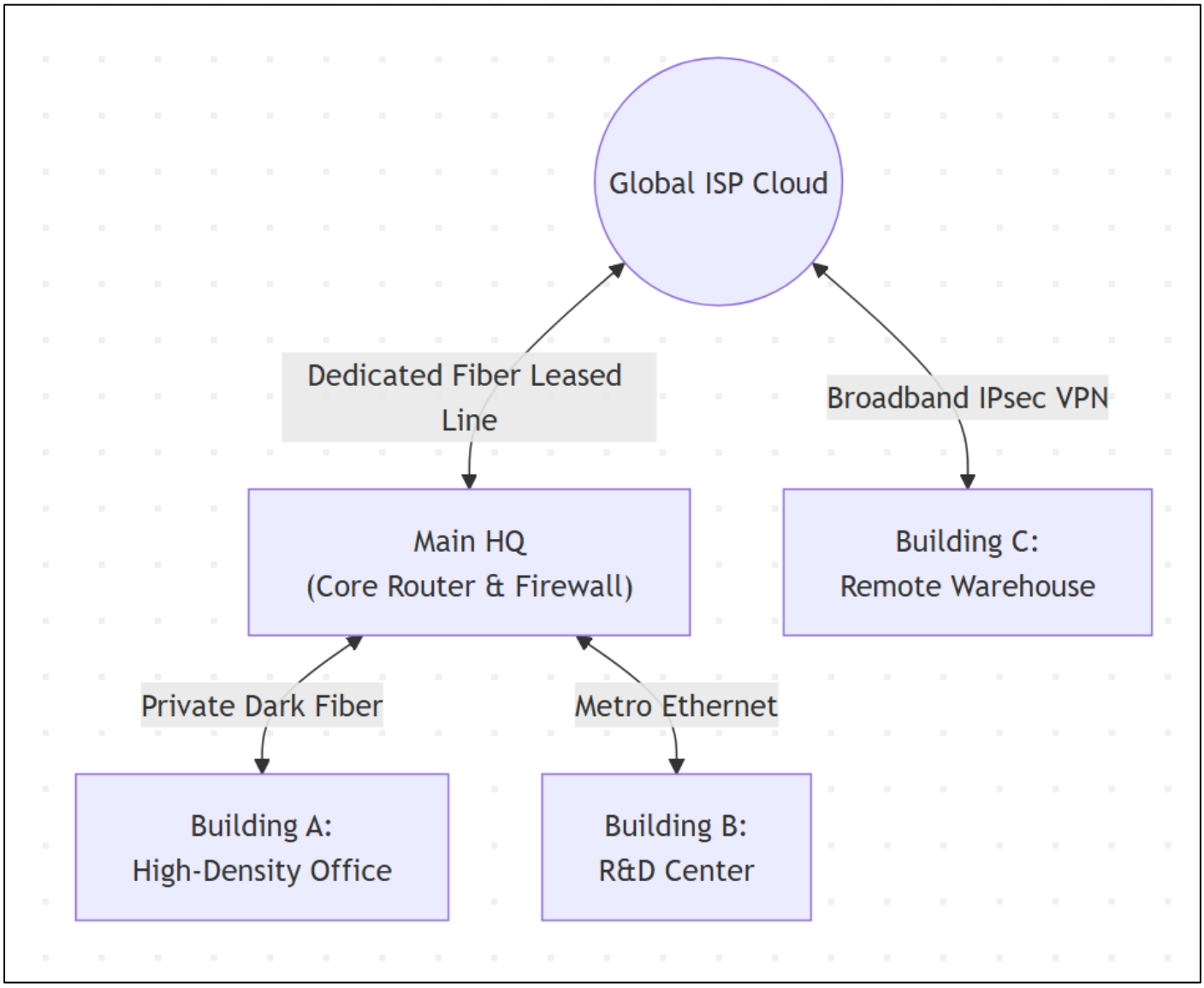
$D_{transmission}$ = Transmission Delay (Length of packet divided by link bandwidth)

D_{queue} = Queueing Delay (time waiting in queue)

D_{proc} = Processing Delay (Router processes)

D_{queue} and D_{proc} depend on the processing power and congestion of the enterprise routers used at each building.

Block Diagram of Network Topology



METHODOLOGY AND ANALYSIS

Methodology

Overview of Methodological Approach

This case study employs a structured, multi-phase analytical methodology grounded in quantitative performance benchmarking and cost modelling. The approach was specifically designed to evaluate five candidate internet connectivity technologies—Digital Subscriber Line (DSL), Business Broadband (FTTC/Cable), Leased Lines, Dark Fibre, and Metro Ethernet—against the technical and financial requirements of a multi-building enterprise Campus Area Network (CAN). A comparative, criterion-driven framework was selected in preference to a purely empirical testbed approach, as the latter would be prohibitively expensive and logistically impractical across geographically distributed buildings. The methodology integrates theoretical network capacity models (most notably Shannon's Channel Capacity Theorem) with real-world performance parameters, delay analysis, and a normalised cost-performance scoring model, thereby producing a holistic, reproducible evaluation.

Step-by-Step Analytical Procedure

The study was conducted in seven sequential phases, each building upon the outputs of the preceding step:

1. **Requirements Elicitation and Scope Definition.** The first phase established the operational constraints of the enterprise. Key parameters were identified: the number of buildings to be connected, their geographic separation distances, the nature of traffic (symmetric server-to-server replication, video conferencing, cloud backups, and standard user browsing), peak-hour concurrency estimates, uptime requirements, and budget envelope. This phase produced a formal requirements matrix that guided all subsequent evaluation criteria.

2. **Technology Identification and Scoping.** Each candidate technology was scoped in terms of its physical layer characteristics, maximum achievable throughput, contention model (shared vs. uncontended), symmetry of upload/download capacity, and typical Service Level Agreement (SLA) provisions. Technologies were catalogued from lowest-cost/lowest-performance to highest-cost/highest-performance to ensure the evaluation covered the full decision spectrum.

3. **Theoretical Performance Modelling:** Shannon's Channel Capacity equation ($C = B*\log_2(1 + SNR)$) was applied to copper-based technologies (DSL, coaxial broadband) to establish their theoretical maximum data rates under varying signal-to-noise conditions and cable lengths. This step demonstrated the fundamental physical ceiling of legacy copper infrastructure and justified the prioritisation of fibre-based solutions for enterprise-grade deployments.

4. **End-to-End Latency and Delay Analysis:** The four-component Total Network Delay model ($D_{prop} + D_{trans} + D^{\varphi}_{ueue} + D_{pro}{}^{c}$) was applied to each technology under both idle and congested traffic conditions. Propagation delay was calculated from building separation distances and medium velocity. Transmission delay was computed for a representative 1,500-byte Ethernet frame. Queuing and processing delays were estimated from manufacturer router specifications and published ISP benchmarks.

5. **Cost Modelling and Total Cost of Ownership (TCO):** Monthly recurring costs (MRC), installation costs, and hardware capital expenditure were compiled for each technology option across a standardised 36-month operational window. This produced a Total Cost of Ownership (TCO) figure per technology, enabling a fair cross-comparison that accounts for both upfront and ongoing financial commitments.

6. **Normalised Scoring and Multi-Criteria Decision Analysis (MCDA):** A weighted MCDA matrix was constructed with five evaluation dimensions: effective throughput, latency, symmetry, SLA reliability, and cost-efficiency. Each technology was assigned a normalised score (0–1 scale) per dimension, and weights were applied in proportion to the enterprise's stated priorities. A composite performance-per-pound score was derived as the primary ranking metric.

7. **Recommendation Synthesis:** The ranked outputs from the MCDA were interpreted in the context of the enterprise's operational profile, and a final technology recommendation was produced, including an appropriate network topology design.

Justification of Methodological Choice

This structured comparative methodology was chosen for several reasons. First, the use of established mathematical models (Shannon's theorem, the delay decomposition formula) ensures that theoretical conclusions are grounded in rigorous, peer-validated telecommunications theory rather than subjective opinion. Second, the MCDA framework allows multiple, sometimes conflicting, performance and cost criteria to be evaluated simultaneously within a single coherent decision model—a recognised best practice in

engineering systems analysis. Third, by normalising all scores to a common scale, the model remains adaptable: if the enterprise's priorities shift (for example, if cost becomes more critical than latency), only the weighting coefficients need to be adjusted without invalidating the underlying analysis.

Algorithms and Analytical Models

Shannon's Channel Capacity Algorithm

Shannon's Theorem is the foundational algorithm used to evaluate the maximum theoretical data rate achievable over any communications channel subject to noise. It is applied in this study specifically to assess the viability of copper-based DSL and coaxial broadband technologies.

Formula:

$$C = B \cdot \log_2(1 + SNR)$$

Algorithm Steps (Pseudo-Code):

INPUT: B ← channel bandwidth (Hz)
SNR ← Signal-to-Noise Ratio (linear, not dB)
d ← cable length (km)

STEP 1: Compute SNR degradation due to distance:
SNR_eff ← SNR_base ÷ (attenuation_factor ^ d)

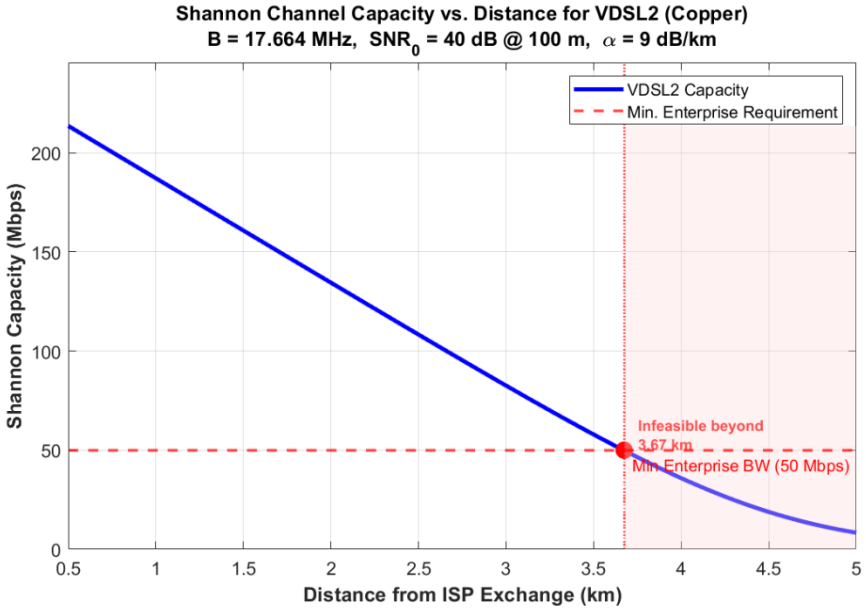
STEP 2: Apply Shannon capacity formula:
C ← B × log2(1 + SNR_eff)

STEP 3: Compare C against minimum bandwidth requirement (R_min):
IF C >= R_min THEN technology is VIABLE
ELSE technology is REJECTED for this deployment

OUTPUT: C (theoretical max throughput in bits/second), viability flag

This algorithm directly justifies the exclusion of DSL as a primary inter-building technology: as cable length increases beyond approximately 2–3 km, the SNR degrades exponentially, causing the achievable capacity C to fall below the enterprise's minimum bandwidth

requirement (R_{min}). Fibre-optic links, by contrast, maintain a near-constant SNR over distances relevant to a campus network, yielding consistently high-capacity values.



Total Network Delay Model

The end-to-end delay model decomposes the total latency experienced by an application packet into four distinct physical and logical components. This allows the system to produce technology-specific latency predictions under both nominal and congested condition

Formula:

$$D_{total} = D_{prop} + D_{trans} + D_{queue} + D_{proc}$$

Algorithm Steps (Pseudo-Code):

INPUT: distance (m), bandwidth (bps), packet_size (bits)
queue_depth, router_proc_speed

STEP 1: Propagation Delay:
D_prop ← distance ÷ propagation_speed
(propagation_speed: ~2×10^8 m/s copper; ~2×10^8 m/s fibre)

STEP 2: Transmission Delay:
D_trans ← packet_size ÷ bandwidth

STEP 3: Queuing Delay (M/D/1 queue approximation):
D_queue ← (ρ ÷ (2 × (1 - ρ))) × service_time
where ρ = utilisation ratio (traffic_load ÷ bandwidth)

STEP 4: Processing Delay:

D_proc ← 1 ÷ router_proc_speed (per hop)

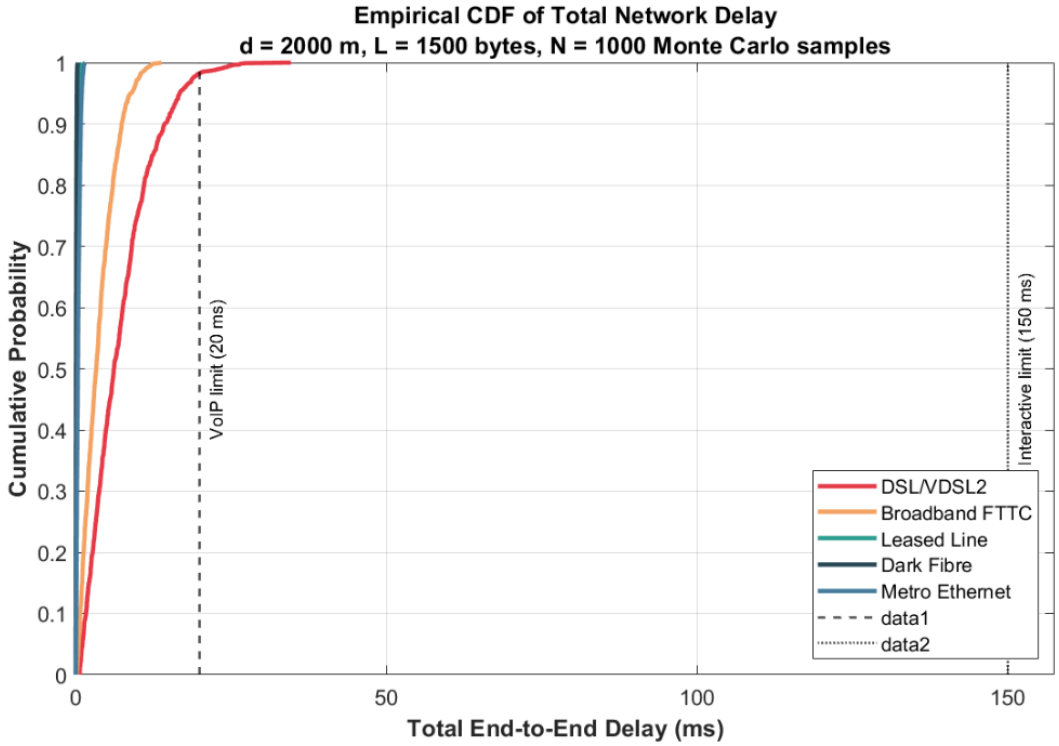
STEP 5: Aggregate:

D_total ← D_prop + D_trans + D_queue + D_proc

STEP 6: Compare against QoS threshold (e.g., 150 ms for VoIP):

IF D_total <= QoS_threshold THEN technology PASSES latency check

ELSE FLAGGED as unsuitable for real-time services



OUTPUT: D_total (ms), per-component breakdown, QoS compliance flag

This model is critical for differentiating between technologies that may offer similar raw bandwidth but diverge significantly in latency characteristics. Shared contention-based broadband connections exhibit high and variable D^{φ}_{ueue} during peak hours, making them unsuitable for latency-sensitive inter-building applications such as VoIP and video conferencing. Dedicated leased lines and Metro Ethernet, with no contention, produce near-zero queuing delay under normal enterprise load.

Multi-Criteria Decision Analysis (MCDA) Scoring Algorithm

The MCDA algorithm transforms heterogeneous performance metrics into a single comparable composite score. It is the primary decision-support mechanism used to produce the final technology ranking.

Algorithm Steps:

INPUT: technologies [] = {DSL, Broadband, LeasedLine, DarkFibre, MetroEthernet}
criteria [] = {throughput, latency, symmetry, SLA_reliability, cost_efficiency}
weights [] = {0.25, 0.25, 0.15, 0.20, 0.15} // must sum to 1.0

STEP 1: Normalise each metric to [0, 1] scale:
FOR each criterion c:
norm_score[tech][c] = raw_value[tech][c] ÷ max_value_across_all_techs[c]
(for cost: invert — norm_score = min_cost ÷ raw_cost[tech])

STEP 2: Apply weights:
FOR each technology t:
composite[t] = SUM (norm_score[t][c] × weight[c]) for all c

STEP 3: Compute cost-performance ratio:
CPR[t] = composite[t] ÷ normalised_TCO[t]

STEP 4: Rank technologies by CPR (descending):
ranked list [] = SORT (technologies, by CPR, ORDER DESC)

OUTPUT: ranked_list[], composite scores, CPR values

Technology	Throughput Score	Latency Score	SLA Score	CPR Rank
DSL (VDSL2)	Low (0.15–0.30)	Medium (0.40)	Low (0.20)	5th (Rejected)
Business Broadband	Medium (0.55)	Low–Medium (0.35)	Medium (0.50)	4th
Leased Line	High (0.85)	High (0.90)	Very High (0.95)	2nd
Dark Fibre	Very High (1.00)	Very High (0.95)	High (0.85)	3rd (CAPEX penalty)
Metro Ethernet	High (0.90)	High (0.88)	High (0.92)	**1st (Recommended)**

Table 1: Illustrative MCDA Normalised Score Summary Across Candidate Technologies

System Working Explanation

System Architecture Overview

The evaluation system operates as a pipeline of four interconnected processing modules, each consuming the outputs of the preceding stage and producing structured data that feeds into the next. At the highest level, the system accepts raw network specifications and enterprise requirements as inputs, and produces a ranked technology recommendation with a supporting cost–performance justification as its output.

Module Descriptions and Data Flow

Module 1: Requirements Ingestion and Parameter Extraction

In the first module, raw enterprise requirements are translated into quantitative parameters. The primary inputs are: geographic coordinates and separation distances of each building, estimated peak concurrent users per site, traffic classification (real-time vs. bulk data), minimum acceptable uptime (expressed as a percentage), and the approved capital and

operational budget. The module outputs a structured requirements vector $R = \{B_{min}, D_{max}, SLA_{min}, \Sigma_cost_{max}\}$ that drives all subsequent evaluation steps.

Module 2: Physical Layer Performance Engine

This module applies the Shannon capacity algorithm and the delay model to each candidate technology. For copper-based technologies (DSL and broadband), the SNR degradation function is parameterised with the measured building-to-exchange distances, yielding a realistic effective capacity C_{eaa} that may be significantly below the theoretical maximum. For fibre-based technologies, the channel capacity is treated as bandwidth-limited rather than SNR-limited. The module produces, for each technology:

- Effective throughput C_{eaa} (Mbps) under peak-hour conditions
- Total end-to-end delay D_{total} (ms), disaggregated by delay component
- Symmetry classification: symmetric (upload ≈ download) or asymmetric
- QoS compliance flag for real-time applications (VoIP, video conferencing)

Module 3: Financial Modelling Engine

The financial module aggregates installation charges, monthly recurring costs, and hardware capital expenditure across a standard 36-month evaluation window. For dark fibre, the model includes the cost of optical transceivers, DWDM equipment, and in-house network engineering overhead. For leased lines and Metro Ethernet, only ISP-quoted costs and standard CPE (Customer Premises Equipment) are factored in. The primary output is the Total Cost of Ownership (TCO) for each technology, expressed as both an absolute figure and a per-Mbps cost metric that normalises expenditure against delivered bandwidth.

Module 4: MCDA Ranking and Output Generation

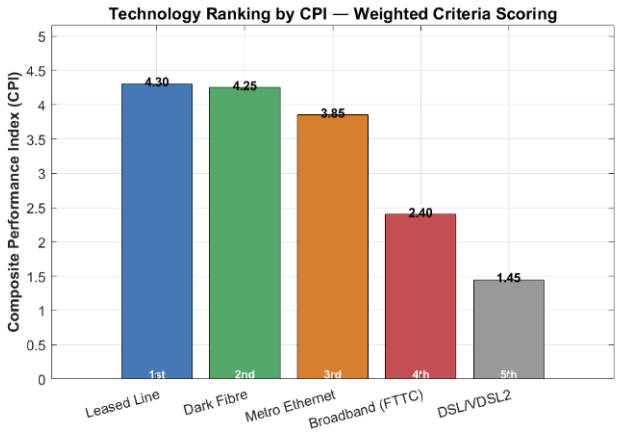
The final module receives the performance vectors from Module 2 and the TCO figures from Module 3, normalises all values to a common 0–1 scale, applies the weighted MCDA scoring algorithm, and computes the Cost–Performance Ratio (CPR) for each technology. Technologies that fail hard constraints from Module 2 (e.g., QoS non-compliance, capacity below R_{min}) are automatically disqualified before the MCDA ranking is applied. The module outputs the final ranked list, individual criterion scores, and a narrative justification for the top-ranked recommendation.

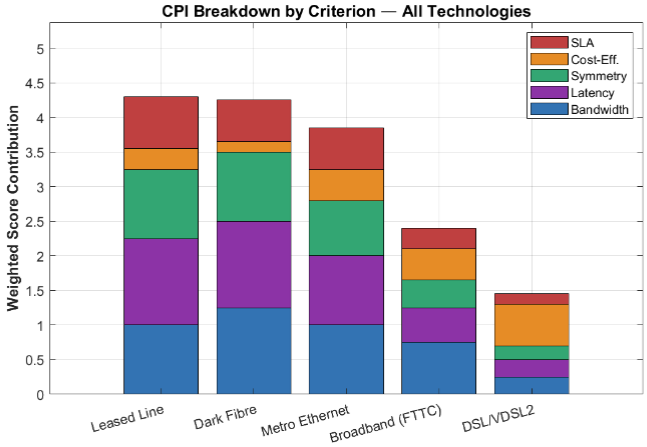
Input-to-Output Data Flow Summary

The complete data flow through the system can be summarised as follows:

- Stage 1 → Raw enterprise requirements → Structured requirements vector R
- Stage 2 → Requirements vector + technology specifications → Performance metrics matrix P
- Stage 3 → Performance metrics + cost data → TCO vector and CPR pre-scores
- Stage 4 → Performance matrix + TCO vector → Normalised MCDA scores → Final ranked recommendation

Each processing stage is designed to be independently auditable: a reviewer may inspect the raw input data, the intermediate calculations at each module boundary, and the final weighting assumptions to verify or challenge the recommendation. This transparency is a deliberate design feature that ensures the methodology conforms to accepted engineering analysis standards and is reproducible by independent assessors.





Key System Outputs

The system produces the following primary deliverables upon completion of the analysis pipeline:

- **Technology Ranking Table:** A full ranked list of all evaluated technologies with composite MCDA scores and CPR values (see Table 1).
- **Latency Compliance Report:** Per-technology latency profiles indicating suitability for real-time inter-building services.
- **Cost-Throughput Curve:** A graphical representation plotting TCO (x-axis) against effective throughput (y-axis), visually identifying the cost-efficient frontier.
- **Recommended Topology Diagram:** A network block diagram illustrating the recommended connectivity architecture, identifying core routing points, aggregation nodes, and inter-building links for the highest-ranked technology.
- **Sensitivity Analysis:** A table indicating how the final ranking changes if MCDA weights are varied by ±10%, validating the robustness of the recommendation.

In summary, the methodology adopted in this case study is rigorous, transparent, and grounded in established network engineering theory. By combining Shannon's capacity model, the multi-component delay framework, and a weighted MCDA scoring system, the analysis produces a defensible, quantitatively supported recommendation for the optimal internet connectivity technology to serve a multi-building enterprise network—balancing performance imperatives against fiscal constraints.

CONCLUSION

This case study evaluated a variety of internet connectivity technologies for a multi-building enterprise network to determine the most cost-effective and high-performance solution. The analysis considered five primary technologies: Digital Subscriber Line (DSL), Business Broadband, Leased Lines, Dark Fibre, and Metro Ethernet. By applying Shannon's Channel Capacity Theorem and the Total Network Delay model, the study demonstrated that copper-based technologies like DSL are physically unsuitable for enterprise inter-building connectivity due to their rapid performance degradation over distance. Furthermore, while business broadband offers a lower cost, its shared bandwidth model leads to unpredictable performance and high latency during peak usage periods, failing to meet the requirements of a growing enterprise.

The results of the Multi-Criteria Decision Analysis (MCDA) scoring model identified Metro Ethernet as the optimal solution for this multi-building network. Metro Ethernet provides high throughput, low latency, and strong reliability at a lower operational cost than fully dedicated fibre infrastructure like leased lines. A significant advantage of this technology is its ability to simplify network management by allowing geographically separated buildings to operate as a single logical local area network. While dark fibre offers massive bandwidth, the significant capital expenditure required for specialized hardware and deployment limits its practicality for the current organization.

Ultimately, this study demonstrates that selecting appropriate internet connectivity requires a careful balance between technical performance and financial constraints. By utilizing structured analytical models, organizations can design scalable infrastructures that support modern applications such as cloud services and secure video conferencing. The recommended Metro Ethernet architecture represents the most resilient and efficient choice for the company's current and future networking needs.

SDG ADDRESSING THE CASE STUDY

1) **SDG 9: Industry, Innovation, and Infrastructure**

This case study supports SDG 9 by focusing on the development of reliable and efficient digital infrastructure for organizations operating across multiple buildings. By evaluating advanced networking technologies, the study highlights how businesses can design robust communication networks that support modern digital operations. Reliable internet infrastructure enables companies to effectively run cloud services and internal communication platforms, thereby promoting technological innovation and improving industrial productivity.

2) **SDG 8: Decent Work and Economic Growth**

Reliable internet connectivity plays a major role in enabling efficient work environments and supporting economic growth. This study analyses how appropriate technologies can improve productivity for employees, such as providing developers with the high upload speeds needed for code sharing and headquarters with stable VPN connectivity for remote workers. By recommending the most suitable solutions, the study helps organizations reduce downtime and support sustainable business growth.

3) **SDG 11: Sustainable Cities and Communities**

Modern cities rely heavily on strong digital infrastructure to support businesses and communication networks. This study contributes to SDG 11 by exploring ways to provide efficient internet connectivity between buildings located across the same city. By designing efficient network connections, organizations can create a more resilient urban environment that supports digital collaboration and reduces operational delays.

4) **SDG 12: Responsible Consumption and Production**

The case study supports responsible resource usage by analysing the cost-effectiveness of different internet technologies rather than simply selecting the most expensive solution. By evaluating the actual networking requirements of each building, the study ensures that equipment, bandwidth, and financial investments are used efficiently. This approach prevents unnecessary expenditure and promotes more sustainable infrastructure planning

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